

Covariance and Covariance Matrix — Solved Problems in Data Science

1. What is Covariance?

Covariance tells us **how two variables change together**.

For two variables X and Y :

Population covariance

$$\text{Cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (x_i - \mu_X)(y_i - \mu_Y)$$

Sample covariance

For real-world data samples, we commonly use:

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y})$$

Interpretation

$$\text{Cov}(X, Y) > 0$$

→ both tend to increase together.

$$\text{Cov}(X, Y) < 0$$

→ one tends to increase while the other decreases.

$$\text{Cov}(X, Y) \approx 0$$

→ little or no **linear** relationship.

Important: Covariance close to zero does not necessarily mean there is no relationship. There could be a nonlinear relationship.

Problem 1 — Study Hours and Exam Score

Data Science Context

Suppose a Data Scientist wants to investigate whether **study hours** are related to **exam scores**.

Student Study Hours X Exam Score Y

1	1	50
2	2	55
3	3	65
4	4	70
5	5	80

Find the covariance between study hours and exam score.

Step 1: Calculate the means

For X:

$$\bar{X} = \frac{1 + 2 + 3 + 4 + 5}{5}$$

$$\bar{X} = 3$$

For Y:

$$\bar{Y} = \frac{50 + 55 + 65 + 70 + 80}{5}$$

$$\bar{Y} = 64$$

Step 2: Calculate deviations

X Y X - $\bar{X}$ Y - $\bar{Y}$ Product

1	50	-2	-14	28
2	55	-1	-9	9
3	65	0	1	0
4	70	1	6	6
5	80	2	16	32

Therefore:

$$\sum (X - \bar{X})(Y - \bar{Y}) = 28 + 9 + 0 + 6 + 32$$
$$\sum (X - \bar{X})(Y - \bar{Y}) = 75$$

Step 3: Sample covariance

Because this is a sample of students, use:

$$S_{XY} = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n - 1}$$
$$S_{XY} = \frac{75}{5 - 1}$$
$$S_{XY} = \frac{75}{4}$$
$$S_{XY} = 18.75$$

Interpretation

The covariance is positive:

$$\text{Cov}(X, Y) > 0$$

Therefore:

Students who study more hours tend to have higher exam scores.

Problem 2 — Temperature and Electricity Consumption

Data Science Context

A Data Scientist wants to investigate the relationship between **outside temperature** and **heating energy consumption**.

Day Temperature X Heating Usage Y

1	10	90
2	15	80
3	20	70
4	25	60
5	30	50

Calculate the covariance.

Step 1: Means

$$\bar{X} = \frac{10 + 15 + 20 + 25 + 30}{5}$$

$$\bar{X} = 20$$

$$\bar{Y} = \frac{90 + 80 + 70 + 60 + 50}{5}$$

$$\bar{Y} = 70$$

Step 2: Deviations

<i>X</i>	<i>Y</i>	$X - \bar{X}$	$Y - \bar{Y}$	Product
10	90	-10	20	-200
15	80	-5	10	-50
20	70	0	0	0
25	60	5	-10	-50
30	50	10	-20	-200

Therefore:

$$\sum (X - \bar{X})(Y - \bar{Y}) = -200 - 50 + 0 - 50 - 200$$

$$\sum (X - \bar{X})(Y - \bar{Y}) = -500$$

Sample covariance:

$$S_{XY} = \frac{-500}{5 - 1}$$

$$S_{XY} = -125$$

Interpretation

$$Cov(X, Y) < 0$$

Therefore:

As outside temperature increases, heating energy consumption tends to decrease.

This is an example of **negative covariance**.

Problem 3 — Construct a Covariance Matrix

This is directly related to the covariance matrix shown in your slide.

Data Science Context

Suppose we have three features:

- X_1 = number of hours spent coding
- X_2 = number of coding exercises completed
- X_3 = number of errors

Consider the following simplified data:

Student	X_1	X_2	X_3
1	1	2	5
2	2	4	4
3	3	6	3
4	4	8	2

Find the covariance matrix.

Step 1: Calculate means

For X_1 :

$$\bar{X}_1 = \frac{1 + 2 + 3 + 4}{4}$$

$$\bar{X}_1 = 2.5$$

For X_2 :

$$\bar{X}_2 = \frac{2 + 4 + 6 + 8}{4}$$

$$\bar{X}_2 = 5$$

For X_3 :

$$\bar{X}_3 = \frac{5 + 4 + 3 + 2}{4}$$

$$\bar{X}_3 = 3.5$$

Step 2: Calculate deviations

X_1	X_2	X_3	$X_1 - 2.5$	$X_2 - 5$	$X_3 - 3.5$
1	2	5	-1.5	-3	1.5
2	4	4	-0.5	-1	0.5
3	6	3	0.5	1	-0.5
4	8	2	1.5	3	-1.5

Step 3: Calculate Variance of X_1

$$\begin{aligned} Var(X_1) &= \frac{(-1.5)^2 + (-0.5)^2 + (0.5)^2 + (1.5)^2}{4 - 1} \\ Var(X_1) &= \frac{2.25 + 0.25 + 0.25 + 2.25}{3} \\ Var(X_1) &= \frac{5}{3} \end{aligned}$$

$$Var(X_1) = 1.667$$

Step 4: Variance of X_2

$$\begin{aligned} Var(X_2) &= \frac{(-3)^2 + (-1)^2 + (1)^2 + (3)^2}{4 - 1} \\ Var(X_2) &= \frac{9 + 1 + 1 + 9}{3} \\ Var(X_2) &= \frac{20}{3} \end{aligned}$$

$$Var(X_2) = 6.667$$

Step 5: Variance of X_3

$$\begin{aligned} Var(X_3) &= \frac{(1.5)^2 + (0.5)^2 + (-0.5)^2 + (-1.5)^2}{4 - 1} \\ Var(X_3) &= \frac{5}{3} \end{aligned}$$

$$Var(X_3) = 1.667$$

Step 6: Covariance Between X_1 and X_2

$$Cov(X_1, X_2) = \frac{\sum (X_1 - \bar{X}_1)(X_2 - \bar{X}_2)}{n - 1}$$

Products:

$$(-1.5)(-3)=4.5$$

$$(-0.5)(-1)=0.5$$

$$(0.5)(1)=0.5$$

$$(1.5)(3)=4.5$$

Therefore:

$$Cov(X_1, X_2) = \frac{4.5 + 0.5 + 0.5 + 4.5}{3}$$

$$Cov(X_1, X_2) = \frac{10}{3}$$

$$Cov(X_1, X_2) = 3.333$$

Positive covariance.

Step 7: Covariance Between X_1 and X_3

Products:

$$(-1.5)(1.5)=-2.25$$

$$(-0.5)(0.5)=-0.25$$

$$(0.5)(-0.5)=-0.25$$

$$(1.5)(-1.5)=-2.25$$

Therefore:

$$Cov(X_1, X_3) = \frac{-2.25 - 0.25 - 0.25 - 2.25}{3}$$

$$Cov(X_1, X_3) = \frac{-5}{3}$$

$$Cov(X_1, X_3) = -1.667$$

Negative covariance.

Step 8: Covariance Between X_2 and X_3

Products:

$$(-3)(1.5)=-4.5$$

$$(-1)(0.5)=-0.5$$

$$(1)(-0.5) = -0.5$$

$$(3)(-1.5) = -4.5$$

Therefore:

$$\begin{aligned} Cov(X_2, X_3) &= \frac{-4.5 - 0.5 - 0.5 - 4.5}{3} \\ Cov(X_2, X_3) &= \frac{-10}{3} \end{aligned}$$

$$Cov(X_2, X_3) = -3.333$$

Step 9: Construct the Covariance Matrix

The covariance matrix is:

$$\Sigma = \begin{bmatrix} Var(X_1) & Cov(X_1, X_2) & Cov(X_1, X_3) \\ Cov(X_2, X_1) & Var(X_2) & Cov(X_2, X_3) \\ Cov(X_3, X_1) & Cov(X_3, X_2) & Var(X_3) \end{bmatrix}$$

Substituting the values:

$$\Sigma = \begin{bmatrix} 1.667 & 3.333 & -1.667 \\ 3.333 & 6.667 & -3.333 \\ -1.667 & -3.333 & 1.667 \end{bmatrix}$$

10. How Do We Read This Matrix?

This is extremely important for students.

Look at the **diagonal**:

$$\Sigma_{11} = 1.667$$

This is:

$$Var(X_1)$$

Similarly:

$$\begin{aligned} \Sigma_{22} &= 6.667 = Var(X_2) \\ \Sigma_{33} &= 1.667 = Var(X_3) \end{aligned}$$

Therefore:

Diagonal elements = Variances

Now look at the **off-diagonal elements**.

$$\Sigma_{12} = 3.333$$

means:

$$\text{Cov}(X_1, X_2) = 3.333$$

And:

$$\Sigma_{13} = -1.667$$

means:

$$\text{Cov}(X_1, X_3) = -1.667$$

Therefore:

Off-diagonal elements = Covariances

11. Why Is the Covariance Matrix Symmetric?

Notice:

$$\text{Cov}(X_1, X_2) = \text{Cov}(X_2, X_1)$$

Therefore:

$$\Sigma_{12} = \Sigma_{21}$$

Similarly:

$$\Sigma_{13} = \Sigma_{31}$$

and:

$$\Sigma_{23} = \Sigma_{32}$$

Hence:

$$\Sigma = \Sigma^T$$

The covariance matrix is always **symmetric**.

Problem 4 — Interpret a Given Covariance Matrix

Suppose a Data Scientist obtains:

$$\Sigma = \begin{bmatrix} 25 & 10 & -5 \\ 10 & 16 & -8 \\ -5 & -8 & 9 \end{bmatrix}$$

where:

- X_1 = Age
- X_2 = Income
- X_3 = Debt

Interpret the matrix.

Step 1: Variance of Age

$$Var(X_1) = 25$$

Therefore:

$$SD(X_1) = \sqrt{25} = 5$$

Step 2: Variance of Income

$$Var(X_2) = 16$$

Therefore:

$$SD(X_2) = \sqrt{16} = 4$$

Step 3: Variance of Debt

$$Var(X_3) = 9$$

Therefore:

$$SD(X_3) = \sqrt{9} = 3$$

Step 4: Age and Income

$$Cov(X_1, X_2) = 10$$

Since:

$$10 > 0$$

there is a positive linear tendency.

Higher age tends to be associated with higher income in this dataset.

Step 5: Age and Debt

$$Cov(X_1, X_3) = -5$$

Since:

$$-5 < 0$$

there is a negative linear tendency.

As age increases, debt tends to decrease in this dataset.

Step 6: Income and Debt

$$Cov(X_2, X_3) = -8$$

Therefore:

Higher income tends to be associated with lower debt in this dataset.

Problem 5 — Find Variance and Covariance From a Covariance Matrix

Suppose:

$$\Sigma = \begin{bmatrix} 9 & 6 \\ 6 & 16 \end{bmatrix}$$

where:

X_1 = House Size

and:

X_2 = House Price

Find:

1. Variance of house size
 2. Variance of house price
 3. Covariance
 4. Standard deviations
 5. Interpret the relationship.
-

1. Variance of house size

From the diagonal:

$$Var(X_1) = 9$$

2. Variance of house price

$$Var(X_2) = 16$$

3. Covariance

Off-diagonal:

$$Cov(X_1, X_2) = 6$$

4. Standard deviations

$$\begin{aligned} SD(X_1) &= \sqrt{9} = 3 \\ SD(X_2) &= \sqrt{16} = 4 \end{aligned}$$

5. Interpretation

Since:

$$\text{Cov}(X_1, X_2) = 6 > 0$$

there is a positive relationship.

Larger houses tend to have higher prices.

Problem 6 — Covariance and Feature Selection

Suppose a Data Scientist has three features:

- X_1 = Age
- X_2 = Years of Experience
- X_3 = Salary

The covariance matrix is:

$$\Sigma = \begin{bmatrix} 25 & 20 & 100 \\ 20 & 16 & 80 \\ 100 & 80 & 400 \end{bmatrix}$$

Notice:

$$\text{Cov}(\text{Age}, \text{Experience}) = 20$$

and:

$$\text{Cov}(\text{Age}, \text{Salary}) = 100$$

and:

$$\text{Cov}(\text{Experience}, \text{Salary}) = 80$$

All are positive.

Interpretation

The variables tend to increase together.

However, here's an important Data Science lesson:

Large covariance does not automatically mean strong correlation.

Why?

Because covariance depends on the **units and scale** of the variables.

For example:

- Age → years
- Salary → rupees
- Experience → years

Therefore, comparing covariance values directly can be misleading.

This leads us to **correlation**.

12. Covariance vs Correlation

This is an excellent point to explain after covariance.

Covariance tells us the **direction** of a linear relationship, but its magnitude depends on units.

Correlation standardizes covariance.

$$\rho_{XY} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

Correlation always lies between:

$$-1 \leq \rho \leq 1$$

Example

Suppose:

$$Cov(X,Y)=6$$

$$\begin{aligned}\sigma_X &= 3 \\ \sigma_Y &= 4\end{aligned}$$

Then:

$$\begin{aligned}\rho &= \frac{6}{(3)(4)} \\ \rho &= \frac{6}{12} \\ \rho &= 0.5\end{aligned}$$

So there is a **moderate positive linear relationship**.

13. Why Covariance Matrix Is Important in Data Science

Now connect the mathematics to actual Data Science.

Application 1 — PCA

The covariance matrix is one of the key inputs to **Principal Component Analysis (PCA)**.

PCA examines how features vary together.

For example:

$$\Sigma = \begin{bmatrix} 10 & 8 \\ 8 & 12 \end{bmatrix}$$

The large positive covariance indicates that the two features move together.

PCA can use this information to find a new direction that captures maximum variance.

Conceptually:

Data → Covariance Matrix → Eigenvalues/Eigenvectors → Principal Components

14. Application 2 — Multivariate Normal Distribution

This directly connects to your previous slide.

For multiple variables:

$$X \sim N(\mu, \Sigma)$$

where:

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix}$$

and:

$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

So:

Σ describes the variance and relationships among features

This is why your slide correctly says:

The covariance matrix is the heart of the multivariate Normal distribution.

15. Application 3 — Machine Learning Features

Suppose we have:

$$\begin{aligned} X_1 &= \text{Study Hours} \\ X_2 &= \text{Practice Questions} \\ X_3 &= \text{Exam Score} \end{aligned}$$

A covariance matrix allows the Data Scientist to understand:

- Which features move together?
- Which features move in opposite directions?
- Which features have high variability?
- Which features may contain redundant information?

16. Application 4 — Finance

Suppose a portfolio contains:

- Stock A
- Stock B
- Stock C

The covariance matrix might be:

$$\Sigma = \begin{bmatrix} 0.04 & 0.01 & -0.005 \\ 0.01 & 0.09 & 0.02 \\ -0.005 & 0.02 & 0.16 \end{bmatrix}$$

The diagonal represents individual asset variances.

The off-diagonal elements represent how asset returns move together.

This is important for:

Portfolio risk estimation

17. A Very Important Concept: Covariance Does NOT Mean Causation

Suppose:

$$\text{Cov}(\text{Ice Cream Sales}, \text{Swimming Pool Visits}) > 0$$

Does buying ice cream cause people to visit swimming pools?

Not necessarily!

A third variable could be responsible:

Temperature

When temperature increases:

- Ice cream sales increase
- Swimming pool visits increase

Therefore, covariance indicates **association**, not causation.

Tell students:

Covariance tells us how variables move together; it does not prove that one variable causes the other.

18. Summary Table for Students

Covariance	Meaning
$\text{Cov}(X,Y) > 0$	Variables tend to increase together
$\text{Cov}(X,Y) < 0$	One tends to increase while the other decreases
$\text{Cov}(X,Y) \approx 0$	Little/no linear relationship
Diagonal of covariance matrix	Variance
Off-diagonal	Covariance

	Covariance	Meaning
Matrix		Symmetric
Units		Product of units of X and Y
Used in		PCA, Multivariate Normal, ML, Finance

★ Final Classroom Example

I would finish the topic with this one because it brings **everything together**.

Suppose:

$$\begin{aligned} X_1 &= \text{Study Hours} \\ X_2 &= \text{Practice Questions} \\ X_3 &= \text{Exam Score} \end{aligned}$$

and the covariance matrix is:

$$\Sigma = \begin{bmatrix} 4 & 6 & 8 \\ 6 & 25 & 15 \\ 8 & 15 & 36 \end{bmatrix}$$

Ask students:

Q1. What is the variance of Study Hours?

4

Q2. What is the variance of Practice Questions?

25

Q3. What is the variance of Exam Score?

36

Q4. What is the covariance between Study Hours and Practice Questions?

6

Q5. What is the covariance between Study Hours and Exam Score?

8

Q6. Is the relationship between Study Hours and Exam Score positive or negative?

Since:

$$r > 0$$

Positive

Q7. What does it mean?

Students who study more hours tend to have higher exam scores.

Q8. Is covariance alone enough to compare the strength of relationships?

No

Because covariance depends on the units and scale.

Use **correlation** when you want a standardized measure:

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$



The concept students should leave with

Give them this mental picture:

Variance = How much one feature varies

Covariance = How two features vary together

Covariance Matrix = All variances + all pairwise covariances

And in Data Science:

Dataset → Covariance Matrix → Understand Feature Relationships → PCA / Multivariate Models / ML

One teaching tip: when you present the covariance matrix, spend more time on **what each cell means** than on the matrix formula itself. Once students understand "diagonal = variance, off-diagonal = covariance," the entire matrix becomes much less intimidating.